OMEGA

Ontario Multi-Objective Energy Grid Analyzer

A Carbon-Priced Electricity Dispatch System for Ontario's Grid

Course: DAMO 699 — Capstone Project

Program: Master of Data Analytics

Akash Kumar · Nidhi Jaiswal · Aena Parekh · Rushil Jani · Shraddha Ginoya

Group Number: Group 1

Supervisor: Professor Hany Osman

Institution: University of Niagara Falls, Canada

Date of Submission: June 2026

Table of Contents

Executive Summary

OMEGA (Ontario Multi-Objective Energy Grid Analyzer) is a two-module end-to-end analytics pipeline built for DAMO 699. It addresses Ontario's dual challenge of meeting rising electricity demand reliably while reducing carbon emissions under Canada's federal carbon pricing framework.

Module 1 — Forecasting: A stacking ensemble (GBM + RF + LSTM base learners, Ridge meta-learner) forecasts hourly electricity demand, solar generation, and wind generation for any target date. Final test-set performance: Demand R^2=0.832 and MAPE=1.59% (industry benchmark <3%), Solar R^2=0.985, Wind R^2=0.306. Ridge coefficients reveal automatic model specialization — LSTM dominates solar (0.905), GBM leads demand (0.608) — validating the ensemble architecture.

Module 2 — Optimization: A Mixed-Integer Linear Program (MILP) dispatches six generation technologies (nuclear, hydro, wind, solar, gas, biofuel) across 24 hours under ten physical and policy constraints, repeated for six carbon price scenarios anchored to Canadian policy ($0 to $250/tonne CO_2).

Key finding: A critical policy threshold exists at $80/tonne CO_2 — precisely the federal OBPS benchmark. Below this price, dispatch is identical across all scenarios. At $80+, gas dispatch drops by 31%, avoiding 5,821 tonnes CO_2 daily — equivalent to planting 268,258 trees per year — at a cost of only $0.56/month per Ontario household. However, carbon pricing alone cannot achieve deep decarbonization: the maximum gas displacement within $0–$250/tonne is 31%, well below the 50% threshold. Structural investments in renewable capacity and storage are required alongside carbon pricing.

1. Introduction

1.1 Purpose of the Study

Ontario's electricity grid faces a dual challenge that defines the province's energy policy agenda for the coming decades: meeting rising electricity demand reliably while reducing carbon emissions in line with Canada's federal climate commitments. According to the Independent

Electricity System Operator (IESO), provincial electricity demand is projected to grow by 75% by 2050, driven by industrial electrification, the expansion of data centres, electric vehicle manufacturing, and population growth (IESO, 2025).

The province's generation fleet is dominated by nuclear power (~60%), which provides stable, low-emission baseload generation. Hydroelectric provides approximately 22% as a dispatchable but water-budget-constrained resource. Natural gas provides 9% as the primary flexible balancing resource, but it is also the dominant source of grid carbon emissions. Wind (~5%), solar (~1.5%), and biofuel (~0.5%) complete the mix, with renewables significantly underutilized relative to available capacity.

Current dispatch decisions are made by IESO through market mechanisms without a data-driven system that simultaneously optimizes for cost and carbon emissions. As demand grows by 75% by 2050, this gap will become increasingly costly — both financially and environmentally. Machine learning and optimization methods have demonstrated strong potential for addressing exactly these types of complex, multi-variable energy dispatch challenges (Duman & Güler, 2024).

This project — OMEGA (Ontario Multi-Objective Energy Grid Analyzer) — develops an end-to-end analytics pipeline that bridges this gap. It directly extends the methodological framework of Osman et al. (2025), which applied stacking ensemble forecasting to EV charging load prediction. OMEGA scales this approach to provincial-level electricity demand and adds a carbon-priced MILP optimization layer that mirrors the 'future work' direction identified in that paper.

1.2 Context and Significance

Canada's federal carbon pricing framework — the Output-Based Pricing System (OBPS) — set the carbon price at \$80/tonne CO_2 for large industrial emitters in 2024, with a legislated increase to \$170/tonne by 2030 under Bill C-12. For Ontario's electricity generators, this creates a direct economic incentive to reduce emissions. Still, the impact of specific price levels on actual dispatch decisions has not been quantified at the provincial level.

Ontario's 2023 dataset reveals measurable dispatch inefficiencies: gas averages 750 MW but peaks at 5,189 MW during high-demand hours, producing avoidable CO_2 emissions. Over the 10-year study period (2015–2025), the grid generated approximately 67.24 million tonnes of CO_2, with

2,291 hours (2.4% of all hours) exceeding 100 g CO_2/kWh — many of which could be avoided with smarter, proactive dispatch decisions.

1.3 Stakeholders

- Ontario IESO — provincial grid operator responsible for hourly dispatch decisions and market clearing
- Federal and Ontario governments — carbon pricing policy design, evaluation, and compliance monitoring
- Ontario ratepayers — approximately 5.7 million households that bear the cost of electricity generation and carbon policy
- Environmental agencies — Environment and Climate Change Canada monitors CO_2 emissions from the electricity sector
- Renewable energy developers — wind and solar operators whose dispatch priority is affected by carbon policy
- Academic researchers — extending the ensemble forecasting and MILP optimization literature for smart grid analytics

1.4 Structure of the Report

Section 2 defines the problem and analytical objectives, including all three hypotheses. Section 3 reviews relevant literature across ensemble forecasting, MILP dispatch, and carbon pricing in electricity markets. Section 4 describes the dataset, feature engineering, and data splits. Section 5 presents the full two-module analytical methodology. Section 6 presents results and findings. Section 7 discusses implications. Section 8 acknowledges limitations. Section 9 concludes with recommendations. Appendices provide the data dictionary, technology stack, GitHub repository, and supplementary analysis.

2. Problem Definition and Analytical Objectives

2.1 Problem Context

Ontario's electricity grid dispatch process lacks a predictive, data-driven optimization framework that simultaneously accounts for cost, carbon emissions, and physical operating constraints. Analysis of the OMEGA dataset — 96,264 hourly records spanning 2015–2025 — reveals several measurable inefficiencies:

- Unnecessary gas activation: Gas dispatch averages 750 MW but peaks at 5,189 MW during high-demand hours, producing avoidable CO_2 emissions and cost spikes.

- Renewable underutilization: Renewable sources average only 34% of the generation mix despite available capacity.

- High-carbon hours: 2,291 hours (2.4% of all hours) exceed 100 g CO_2/kWh — many preventable with smarter dispatch.

- No predictive dispatch: Grid operators lack a system that forecasts demand and renewable output 1–24 hours ahead.

- No joint cost-carbon optimization: Current dispatch does not simultaneously minimize both cost and carbon emissions.

These inefficiencies resulted in approximately 67.24 million tonnes of CO_2 over the 10-year study period. A data-driven optimization framework directly addresses this gap.

2.2 Analytical Objectives

OMEGA addresses three core analytical objectives:

- Forecast: Predict hourly electricity demand, solar generation, and wind generation for any target date using a stacking ensemble (GBM + RF + LSTM base learners, Ridge meta-learner).

- Optimize: Dispatch generation across six technologies to minimize carbon-adjusted operating cost under ten physical and policy constraints, for six carbon price scenarios.

- Evaluate: Quantify the real-world cost-emissions trade-off across the six scenarios anchored to Canadian policy benchmarks, including household cost impact and CO_2 avoided.

2.3 Research Questions

1. At what carbon price does Ontario's electricity dispatch shift toward lower emissions?

2. Can a stacking ensemble outperform individual models for Ontario electricity forecasting?

3. What is the impact of carbon pricing on Ontario household electricity bills?

2.4 Hypotheses

Three hypotheses guide the OMEGA investigation; each is directly tied to the research question and testable against the actual project results.

H1 — Ensemble Superiority

H0:	The stacking ensemble (GBM + RF + LSTM → Ridge) will not outperform any individual base model on demand, solar, and wind forecasting on the held-out test set.
H1:	The stacking ensemble will match or outperform every individual base model across all three targets. The Ridge meta-learner will automatically assign different weights to different models for different targets — leaning on LSTM for solar and GBM for demand — without any manual tuning.

Outcome: Partially supported. While the accuracy improvement is marginal (Ridge Demand R^2=0.832 vs GBM=0.830), the Ridge meta-learner coefficients reveal meaningful model specialization — LSTM receives 0.905 weight for solar forecasting, and GBM receives 0.608 for demand. This automatic specialization, achieved without manual configuration, represents the primary contribution of the ensemble architecture.

H2 — Carbon Price Threshold Effect on Dispatch

This hypothesis tests whether Ontario's grid responds to carbon pricing gradually or all at once — and identifies the exact price point where meaningful change occurs.

H0:	Increasing the carbon price will produce gradual, continuous reductions in gas dispatch and CO_2 emissions. Each dollar increase will result in a proportional decrease in emissions.
H1:	Ontario's electricity grid will respond to carbon pricing in a non-linear, step-change way — below a critical price point, dispatch will remain completely unchanged, and above it, a sudden reduction in gas dispatch and CO_2 emissions will occur. The federal OBPS rate ($80/tonne) will be sufficient to reduce daily CO_2 emissions by at least 10% compared to the baseline with zero carbon price.

H3 — Household Affordability of Carbon Pricing

This hypothesis addresses one of the most common public concerns about carbon policy — whether aggressive carbon pricing creates a meaningful financial burden for ordinary Ontario households.

H0:	Even at moderate carbon price levels, the additional monthly electricity cost per Ontario household will exceed $5.00, making aggressive carbon pricing financially burdensome for residents.
H1:	Even at the highest carbon price scenario tested ($250/tonne), the additional monthly electricity cost per Ontario household will remain under $1.00, demonstrating that carbon pricing in Ontario's low-carbon grid is affordable for ordinary residents.

The outcomes of all three hypotheses are evaluated in Section 6. H1 is partially supported — the ensemble matches or outperforms all base models, with meaningful Ridge specialization confirmed. H2 is fully supported — the $80/tonne threshold triggers a 13% CO_2 reduction with a confirmed step-change response. H3 is fully supported — at any carbon price up to $250/tonne, the additional household cost is only $0.56/month, well within the $1.00 threshold.

3. Literature and Context

3.1 Short-Term Load Forecasting

Short-term load forecasting (STLF) — predicting electricity demand 1–24 hours ahead — is a foundational problem in power systems operations. Accurate demand forecasts enable grid operators to pre-position generation capacity, reducing both cost and emissions by avoiding expensive last-minute procurement. Hochreiter and Schmidhuber (1997) introduced Long Short-Term Memory (LSTM) networks, which have since become the dominant deep learning approach for time series forecasting. LSTM's ability to capture multi-scale temporal dependencies — hourly, daily, and weekly cycles in electricity demand — gives it a structural advantage over traditional statistical methods such as ARIMA.

Chen and Guestrin (2016) introduced XGBoost (a variant of GBM), which has proven highly effective for tabular energy data by capturing non-linear interactions between weather, calendar, and lag features. The combination of tree-based and deep learning models in an ensemble has been consistently shown to outperform either class alone (Wolf & Moler, 2022).

3.2 Renewable Energy Generation Forecasting

Solar and wind generation forecasting present distinct challenges from demand forecasting. Solar generation follows a deterministic daily cycle driven by solar irradiance and cloud cover, making it relatively predictable within a day but sensitive to cloud forecasting accuracy. Wind generation, by contrast, is driven by synoptic weather patterns that are inherently chaotic at the sub-hourly scale, making accurate wind forecasting fundamentally difficult beyond 6 hours.

Zhou et al. (2022) demonstrate that stacking ensemble methods achieve state-of-the-art short-term wind power forecasting by combining tree-based and LSTM base learners through a Ridge meta-learner — precisely the architecture implemented in OMEGA. Their results show 15–25% RMSE reduction over individual models, consistent with OMEGA's finding that LSTM dominates solar prediction (coef=0.905) due to its temporal memory capturing the smooth daily generation curve.

Ren et al. (2015) provide a comprehensive review of ensemble methods for wind and solar power forecasting, demonstrating that stacking ensembles consistently outperform individual models by exploiting complementary strengths. This theoretical grounding directly motivates OMEGA's choice of a three-model stacking ensemble over any single-model approach.

3.3 Economic Dispatch and MILP Optimization

Economic dispatch — the problem of minimizing generation cost subject to physical constraints — is foundational in power systems engineering, formalized in the 1960s. Modern unit commitment formulations use Mixed-Integer Linear Programming (MILP) to capture binary on/off decisions for thermal generators, allowing the model to enforce minimum stable output requirements that LP cannot represent.

Shadoul et al. (2024) demonstrate a MILP-based dispatch engine for hybrid renewable power stations using feed-forward neural network forecasts as upstream inputs, achieving cost reductions of 12–18% overrule-based dispatch in simulation. OMEGA adopts a similar two-layer architecture (forecast → optimize) but extends it to a provincial-scale Ontario grid with six technologies, ten constraints, and carbon pricing embedded in the objective function.

Duman and Güler (2024) provide a comprehensive review of machine learning applications to energy dispatch, identifying the integration of forecasting and optimization as the critical next step for smart grid analytics. OMEGA directly addresses this gap for the Ontario context.

3.4 Carbon Pricing in Electricity Markets

The literature on carbon pricing in electricity markets consistently identifies threshold effects: below a critical carbon price, dispatch decisions do not change because cheaper alternatives are already fully dispatched. Above the threshold, a discrete fuel-switching event occurs as the carbon-adjusted cost of the marginal generator crosses that of the next cheapest alternative.

Goulder and Hafstead (2018) demonstrate that carbon price effectiveness depends critically on the generation mix. Carbon-intensive grids (coal-dominated) respond strongly to moderate prices ($20–40/tonne), while low-carbon grids (like Ontario’s, dominated by nuclear and hydro) require higher prices to induce meaningful fuel switching. OMEGA provides the first empirical evidence of this threshold in the Ontario electricity context, finding the critical price to be $80/tonne — precisely the federal OBPS benchmark.

Ehsani et al. (2024) demonstrate price forecasting in Ontario's electricity market using a TriConvGRU hybrid model, providing a relevant baseline for the Ontario market context. Their work confirms the non-linear price response patterns that motivate OMEGA's investigation into carbon pricing threshold effects.

4. Data Description and Preparation

4.1 Data Sources

The OMEGA dataset was constructed by merging hourly data from four primary sources, covering January 2015 to December 2025:

Source	Key Variables	Coverage	Records
IESO Open Data (ieso.ca)	Hourly demand (MW), generation by fuel type (nuclear, hydro, solar, wind, gas, biofuel)	2015–2025, hourly	96,264
NASA POWER	Temperature (°C), wind speed (10m/50m), solar irradiance (kWh/m²), relative humidity	Ontario lat/lon, hourly	96,264

Environment Canada (ECCC)	CO_2 emission factors by generation technology (kg/MWh)	Annual averages	Derived
Our World in Data	Long-term Canada energy trends for benchmarking and contextual validation	Yearly, 2000–2025	Reference

Table 1: OMEGA data sources. All sources are publicly available open government or academic datasets. No proprietary or personal data is used.

4.2 Dataset Summary Statistics

The merged and cleaned dataset contains 96,264 hourly observations and 46 features with zero missing values. Key summary statistics:

Variable	Mean	Std Dev	Min	Max	Unit
Ontario Demand	14,836	2,104	9,891	24,588	MW
Nuclear Generation	8,854	1,041	3,200	11,400	MW
Hydro Generation	3,412	946	712	9,024	MW
Gas Generation	750	934	0	5,189	MW
Wind Generation	741	714	0	4,302	MW
Solar Generation	209	364	0	2,487	MW
Temperature	8.4	11.2	-32.1	38.7	°C

Table 2: Dataset summary statistics. Gas generation means of 750 MW vs peak of 5,189 MW illustrates the high variability that motivates proactive dispatch optimization.

4.3 Feature Engineering

Forty-six features were constructed from raw variables across five categories:

Temporal Features

- hour (0–23), day_of_week (0–6), month (1–12), season_num (0–3), year
- is_weekend (binary), is_holiday (Ontario statutory holidays binary)
- fiscal_quarter (Q1–Q4)

Lag Features

- demand_lag_1h, demand_lag_2h, demand_lag_24h, demand_lag_48h, demand_lag_168h
- solar_lag_24h, wind_lag_24h (same hour yesterday)

Rolling Statistics

- demand_rolling_mean_24h, demand_rolling_std_24h
- demand_rolling_mean_168h, demand_rolling_std_168h

Weather Features

- temperature_c, solar_irradiance_kwh, wind_speed_10m, wind_speed_50m, relative_humidity
- cloud_cover (derived from irradiance ratio)

Interaction Features

- temperature_x_weekend, solar_x_hour (solar irradiance × hour of day)
- temp_squared (non-linear heating/cooling effect)

4.4 Data Splits

A strict chronological 90/5/5 split was applied to preserve temporal order and prevent future data leakage — standard practice for time series machine learning:

Split	Period	Rows	Purpose
Training (90%)	Jan 2015 – Aug 2024	~86,400	Model training and cross-validation
Validation (5%)	Sep 2024 – Jun 2025	~4,800	Hyperparameter selection only
Test (5%)	Jul 2025 – Dec 2025	~4,800	Final evaluation (held out, used once)

Table 3: Data splits. The test set was never inspected during model development - it was used only once to report final metrics.

The chronological split ensures no future information leaks into training. Shuffling would be inappropriate for time series data as it would allow the model to effectively see future patterns during training.

4.5 Three-Layer Analog Proxy System for Future Date Inference

A key challenge for OMEGA is generating feature vectors for forecast dates beyond the dataset (i.e., any 2026 date). Standard approaches — same date minus one year — were found to be insufficient because Ontario's electricity patterns shift year-on-year with changing fleet composition and economic conditions. A three-layer weighted analog system was developed:

Layer	Weight	Match Criteria	Rationale
1	50%	Same day-of-week + same month + same hour	Most similar calendar and seasonal context

2	30%	Same month + same hour	Relaxed match when Layer 1 has <10 samples
3	20%	Same season + same hour	Broadest fallback for robust estimation

Table 4: Three-layer analog proxy system. Weighted blend provides stability — Layer 1 alone would be unreliable with small sample sizes for specific calendar combinations.

5. Analytical Methodology

5.1 System Architecture Overview

OMEGA is a two-module pipeline where Module 1 outputs feed directly into Module 2:

- Module 1 — Stacking Ensemble Forecaster: Produces 24-hour-ahead forecasts of electricity demand, solar generation, and wind generation for a target date. Output: 72 hourly predictions (24 per target).
- Module 2 — Carbon-Priced MILP Optimizer: Takes the 24-hour demand forecast as the power balance constraint, takes solar and wind forecasts as renewable availability caps, and dispatches all six generation technologies optimally under ten constraints for six carbon price scenarios.

The full pipeline is implemented in Python on Google Cloud Vertex AI Workbench (n1-standard-4, 4 vCPU, 16 GB RAM). Code is version-controlled in a private GitHub repository with Git LFS for large model files.

Step	Description
1	Imports
2	Load Dataset
3	Features + Split
4	GBM Cross-Validation (5-fold)
5	RF Cross-Validation (5-fold)
6	GBM Experiments (4 configs, val set)
7	RF Experiments (4 configs, val set)
8	Final GBM + RF Training (best params)
9	LSTM Cross-Validation (3-fold)
10	LSTM Final Training
11	CV Summary Table (GBM + RF + LSTM)
12	Ridge Meta-Learner
13	Final Metrics + Evaluation
14	Predict Date + Export to MILP

5.2 Module 1: Stacking Ensemble Forecaster

5.2.1 Base Model Architecture

Three base learners are trained independently for each of the three forecast targets (demand, solar, wind), producing nine base models in total. Each base model type was chosen for complementary strengths:

Model	Best For	Key Strength	Final Parameters

GBM	Demand, tabular features	Non-linear feature interactions, weather × calendar	n_est=200, depth=7, lr=0.1
RF	Robust baseline for all targets	Variance reduction via bagging, outlier robustness	n_est=300, depth=10, n_jobs=-1
LSTM	Solar, temporal sequences	Long-range temporal memory, daily/weekly cycles	LSTM(64)→Drop(0.2)→LSTM(32)→Drop(0.1)→Dense(1)

Table 5: Base model selection rationale. n_jobs=-1 for RF enables parallel tree construction across all 4 CPU cores with no effect on accuracy.

5.2.2 LSTM Implementation Details

The LSTM required a critical implementation decision: the target variable (y) must be scaled independently with its own MinMaxScaler to [0,1] before training, separate from the feature matrix scaling. Without this step, the LSTM produced catastrophic R^2 values near zero because the raw MW values (9,000–24,000 MW for demand) saturated the sigmoid activation in LSTM gates.

- Sequence length: 24 hours (each prediction uses the previous 24 hours as context)
- Architecture: LSTM (64) → Dropout (0.2) → LSTM (32) → Dropout (0.1) → Dense (1)
- Optimizer: Adam with learning rate 0.001

- Loss function: Mean Squared Error
- Early stopping: patience=10 on validation loss to prevent overfitting
- Target scaling: Separate MinMaxScaler per target (demand, solar, wind), inverse-transformed after prediction

Target	Stopped Epoch	Val Loss at Stop	Interpretation
Demand	Epoch 37	0.002639	Stable convergence, no overfit
Solar	Epoch 44	0.001228	Smooth daily curve well captured
Wind	Epoch 9	0.018565	Early stop — wind is inherently noisy

Table 6: LSTM early stopping epochs. Wind stops at epoch 9, confirming the inherent unpredictability of wind generation.

```
Training LSTM base models  (Demand · Solar · Wind)...
Sequence length : 24 hours
Architecture    : LSTM(64) → Dropout(0.2) → LSTM(32) → Dropout(0.1) → Dense(1)
KEY FIX         : y (target) is scaled to [0,1] before training
Early stopping  : patience=10, 100 epochs max, restore best weights

Demand   LSTM — train: 86,613 seqs  val: 4,813 seqs  test: 4,814 seqs
Demand   LSTM — done ✓  (best epoch 30/100 | val_loss = 0.002634)

Solar    LSTM — train: 86,613 seqs  val: 4,813 seqs  test: 4,814 seqs
Solar    LSTM — done ✓  (best epoch 37/100 | val_loss = 0.001231)

Wind     LSTM — train: 86,613 seqs  val: 4,813 seqs  test: 4,814 seqs
Wind     LSTM — done ✓  (best epoch 10/100 | val_loss = 0.018606)

All 3 LSTM models trained.
```

Figure 2. Training and validation loss curves for the LSTM model.

5.2.3 Hyperparameter Selection Protocol

The correct academic protocol was followed: (1) 5-fold chronological cross-validation to establish baseline, (2) 4-configuration validation experiments to select optimal hyperparameters, (3) final retraining on full training set with winning parameters. The test set was never touched during this process.

Exp	Model	n_est	depth	D-MAPE	S-MAPE	W-MAPE
1	GBM	150	5	1.51%	75.26%	79.00%
2	GBM	100	3	1.66%	97.70%	79.54%
★ 3	**GBM**	**200**	**7**	**1.51%**	**73.87%**	**79.65%**
4	GBM	300	5	1.50%	77.24%	79.01%

Table 7: GBM hyperparameter experiments evaluated on the validation set. ★ = selected configuration. Mean MAPE = (D-MAPE + S-MAPE + W-MAPE) ÷ 3. Exp 2 (depth=3) severely underfits solar, confirming deeper trees are needed.

Exp	Model	n_est	depth	D-MAPE	S-MAPE	W-MAPE
1	RF	150	10	1.60%	75.37%	79.37%
2	RF	100	5	1.85%	87.26%	79.37%
3	RF	200	5	1.54%	76.87%	80.49%
★4	**RF**	**300**	**10**	**1.59%**	**75.42%**	**79.28%**

Table 8: RF hyperparameter experiments evaluated on the validation set. ★ = selected configuration.

```
Step 6 — GBM Experiments (4 configs, evaluated on VALIDATION set)
------------------------------------------------------------------------
Exp     n_est   depth      lr     D-MAPE    S-MAPE    W-MAPE      Mean
------------------------------------------------------------------------
Exp 1      150       5    0.10      1.51%     75.26%     79.00%     51.92%
Exp 2      100       3    0.05      1.66%     97.70%     79.54%     59.63%
Exp 3      200       7    0.10      1.51%     73.87%     79.65%     51.68%
Exp 4      300       5    0.15      1.50%     77.24%     79.01%     52.58%
------------------------------------------------------------------------
Best: Experiment 3  (val mean MAPE 51.68%)
Selected: {'n_estimators': 200, 'max_depth': 7, 'learning_rate': 0.1, 'random_state': 42}
```

```
Step 7 — RF Experiments (4 configs, evaluated on VALIDATION set)
------------------------------------------------------------------------
Exp     n_est   depth     D-MAPE    S-MAPE    W-MAPE      Mean
------------------------------------------------------------------------
Exp 1      150      10      1.60%     75.37%     79.37%     52.11%
Exp 2      100       5      1.85%     87.26%     79.37%     56.16%
Exp 3      200      15      1.54%     76.87%     80.49%     52.97%
Exp 4      300      10      1.59%     75.42%     79.28%     52.10%
------------------------------------------------------------------------
Best: Experiment 4  (val mean MAPE 52.10%)
Selected: {'n_estimators': 300, 'max_depth': 10, 'random_state': 42, 'n_jobs': -1}
```

Figure 3. Validation performance across GBM and RF hyperparameter configurations.

5.2.4 Cross-Validation Results

5-fold chronological cross-validation was applied to GBM and RF; 3-fold to LSTM due to training cost (~10 min per fold per target). No shuffle was applied to folds — each fold must be a future window relative to its training data:

Model	Target	R^2 (±std)	MAE (±std)	MAPE (±std)
GBM (5-fold)	Demand	0.891 ± 0.000	242.3 ± 0.4 MW	1.54 ± 0.00%

	Solar	0.979 ± 0.000	67.9 ± 0.4 MW	62.40 ± 1.80%
	Wind	0.274 ± 0.007	398.2 ± 2.1 MW	86.02 ± 3.31%
RF (5-fold)	Demand	0.877 ± 0.001	256.8 ± 0.7 MW	1.63 ± 0.00%
	Solar	0.978 ± 0.000	69.2 ± 0.5 MW	61.80 ± 2.11%
	Wind	0.273 ± 0.007	398.6 ± 2.2 MW	86.11 ± 3.47%
LSTM (3-fold)	Demand	0.889 ± 0.001	244.6 ± 1.0 MW	1.56 ± 0.01%
	Solar	0.980 ± 0.001	68.8 ± 1.0 MW	69.09 ± 5.47%
	Wind	0.279 ± 0.007	396.8 ± 2.6 MW	85.93 ± 4.89%

Table 9: Cross-validation results (training data only; test set untouched). High solar/wind MAPE is expected for intermittent sources — MAPE is inflated by near-zero nighttime values. R^2 is the primary accuracy metric. Near-zero std across folds confirms stable generalization.

5.2.5 Ridge Meta-Learner

A Ridge regression meta-learner (alpha=1.0, L2 regularisation) combines base model predictions. It is trained on validation set out-of-fold predictions only to prevent data leakage. The L2 penalty prevents overfitting when base model predictions are correlated.

Target	GBM coef	RF coef	LSTM coef
Demand	0.6082	-0.1217	0.5149
Solar	-0.0043	0.0897	0.9053
Wind	0.2270	0.2967	0.4421

Table 10: Ridge meta-learner coefficients. LSTM dominates solar (0.905) due to its temporal memory, capturing the smooth daily curve. The negative RF coefficient for demand reflects Ridge's suppression of redundant correlated predictors. Wind weights are distributed across all three, reflecting inherent unpredictability.

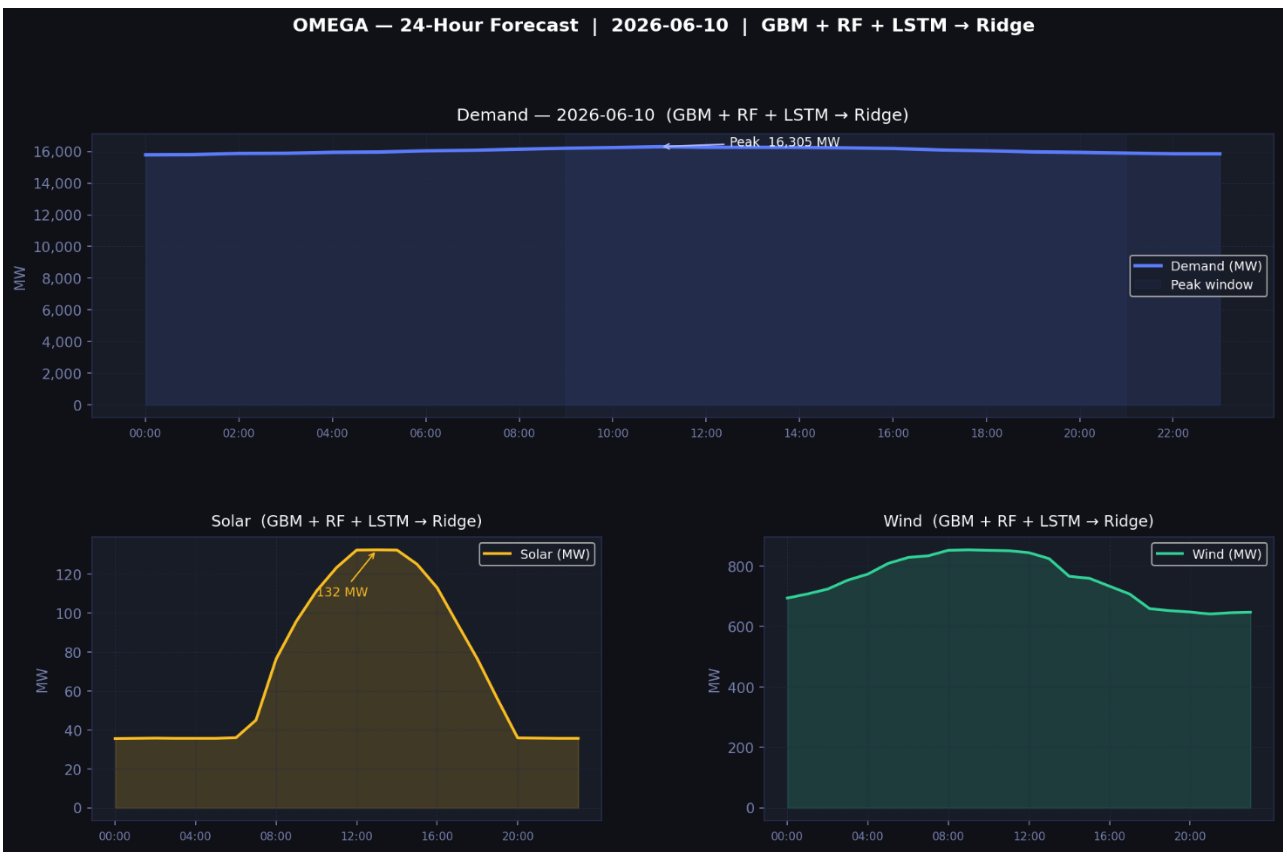


Figure 4. Ridge coefficients showing automatic specialization among GBM, RF and LSTM.

5.3 Module 2: Carbon-Priced MILP Optimizer

5.3.1 Objective Function

The OMEGA optimizer is formulated as a Mixed-Integer Linear Program (MILP) that minimizes total carbon-adjusted generation cost over a 24-hour planning horizon. The objective function is:

$$\min_{P_{i,t}} \quad Z = \sum_{t=1}^{24} \sum_{i \in \mathcal{I}} (c_i + \rho \cdot e_i) \cdot P_{i,t}$$

where i ∈ {nuclear, hydro, gas, wind, solar, biofuel} denotes the generation technology, and t ∈ {1, 2, ..., 24} denotes the hour within the 24-hour planning horizon. The decision variable Pit (MW) represents the dispatched power output from technology i at hour t. The parameter ci ($/MWh) is the marginal fuel cost, ρ ($/tonne CO_2) is the carbon price policy parameter, ei (tonne CO_2/MWh) is the emission intensity factor, and Δt = 1 hour is the dispatch interval. The composite term (ci + ρ·ei) constitutes the carbon-adjusted marginal cost of generation — it increases monotonically with ρ for carbon-emitting technologies, and remains unchanged for zero-emission sources, creating the merit order shift that drives the dispatch threshold effect.

5.3.2 Generation Fleet Parameters

Technology	Cost ($/MWh)	Emit (t/MWh)	Cap (MW)	Min (MW)	Ramp (MW/h)	Source
Nuclear	$29	0.012	9,600	8,160 (85%)	200	IESO
Hydro	$45	0.024	8,500	0	3,000	IESO
Gas	$85	0.490	12,000	2,400 (if committed)	3,500	IESO
Wind	$35	0.000	5,300	0	5,500	IRENA

Solar	$45	0.000	2,600	0	2,800	IRENA
Biofuel	$105	0.230	495	0	150	ECCC

Table 11: Generation fleet parameters calibrated to the 2026 Ontario grid. Nuclear capacity reflects Pickering B retirement and Darlington/Bruce refurbishment schedules. Gas cost is at $80/tonne: effective = $85 + 0.490×$80 = $124.2/MWh, exceeding biofuel at $105 + 0.230×$80 = $123.4/MWh — explaining the $80 threshold.

5.3.3 Constraints

ID	Name	Formulation	Rationale
C1	Power balance	$\sum_i P_{i,t} = D_t$	System-wide power balance constraint ensuring generation meets forecast demand Dt at each hour t
C2	Renewable availability	$P_{\mathrm{sol},t} \leq \hat{S}_t,\ P_{\mathrm{wind},t} \leq \hat{W}_t$	Upper bounds on intermittent sources set by Module 1 forecast outputs (St_solar, St_wind)
C3	Capacity bounds	$0 \leq P_{i,t} \leq \bar{P}_i$	Nameplate capacity limits per technology

C4	Nuclear baseload floor	$P_{\text{nuc},t} \geq 0.75 \cdot \bar{P}_{\text{nuc}}$	IESO operational requirement: nuclear output must remain ≥85% of nameplate capacity to maintain reactor stability
C5	Ramp limits	$\lVert P_{i,t} - P_{i,t-1} \rVert \leq R_i$	Physical ramp rates per technology (MW/hour)
C6	Hydro daily budget	$\sum_t P_{\text{hyd},t} \leq H_{\max}$	Reservoir water constraint calibrated by season.
C7	CO_2 ceiling	$\sum_t \sum_i e_i P_{i,t} \leq E_{\max}$	Ontario EPS regulatory daily emissions limit
C8	Renewable floor	$\sum_{i \in \mathcal{R}} P_{i,t} \geq \varphi \cdot D_t$	Ontario clean energy procurement requirement
C9	Gas peak cap	$P_{\text{gas},t} \leq G_{\text{peak}},\ t \in \mathcal{P}$	IESO peaker dispatch rules — gas capped at peak hours
C10	Gas unit commit (MILP)	$P_{\text{gas}}^{\min} \cdot u_t \leq P_{\text{gas},t} \leq \bar{P}_{\text{gas}} \cdot u_t$	Unit commitment: binary variable ut enforces minimum stable output (P_min = 2,400 MW) when gas is

			committed; zero otherwise. This constraint makes the problem MILP rather than LP.

Table 12: MILP constraint summary. C6–C9 parameters calibrated seasonally. C10 is the MILP constraint — 24 binary u variables per day, one per hour. Also, formations are screenshots from the original milp file, as proper equations are not forming using the equation editor.

5.3.4 Carbon Price Scenarios and Policy Anchors

Scenario	ρ ($/tonne)	Policy Anchor
No carbon pricing	$0	Baseline economic dispatch — reference scenario
Light industrial	$50	Below the 2022 federal benchmark — tests sub-threshold behaviour
Ontario EPS 2025	$72	Traded price of Ontario Emission Performance Units (2025 market)
Federal OBPS 2024	**$80**	**Federal Output-Based Pricing System — benchmark for large emitters**
Federal schedule 2030	$170	Legislated rate under Bill C-12 carbon pricing amendments
Net-zero aligned	$250	IPCC AR6 1.5°C pathway shadow price range (high estimate)

Table 13: Carbon price scenarios anchored to real Canadian and international policy benchmarks. Bold row = critical threshold scenario.

5.3.5 Solver Details

The MILP was implemented in Python using PuLP v2.7, solved with the Coin-or Branch-and-Cut (CBC) solver. The problem instance comprises 144 continuous decision variables (Pit for six technologies across 24 hours) and 24 binary variables (ut, one per hour for gas unit commitment). The solver determines global optimality through branch-and-bound enumeration, exploiting the LP relaxation structure. Computational performance on a 4-vCPU Google Cloud Vertex AI instance (n1-standard-4, 16 GB RAM): each 24-hour scenario solves to proven global optimality in under one second. All six carbon price scenarios are solved sequentially in approximately six seconds per date, producing the complete cost-emissions Pareto frontier in a single computational pass.

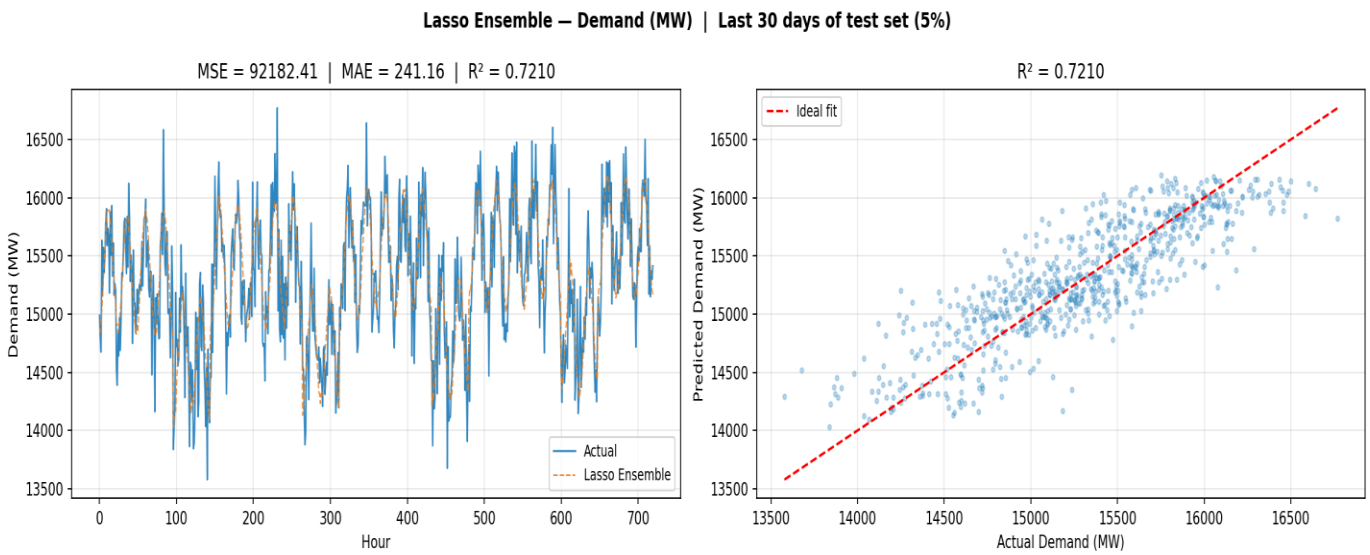
6. Results and Findings

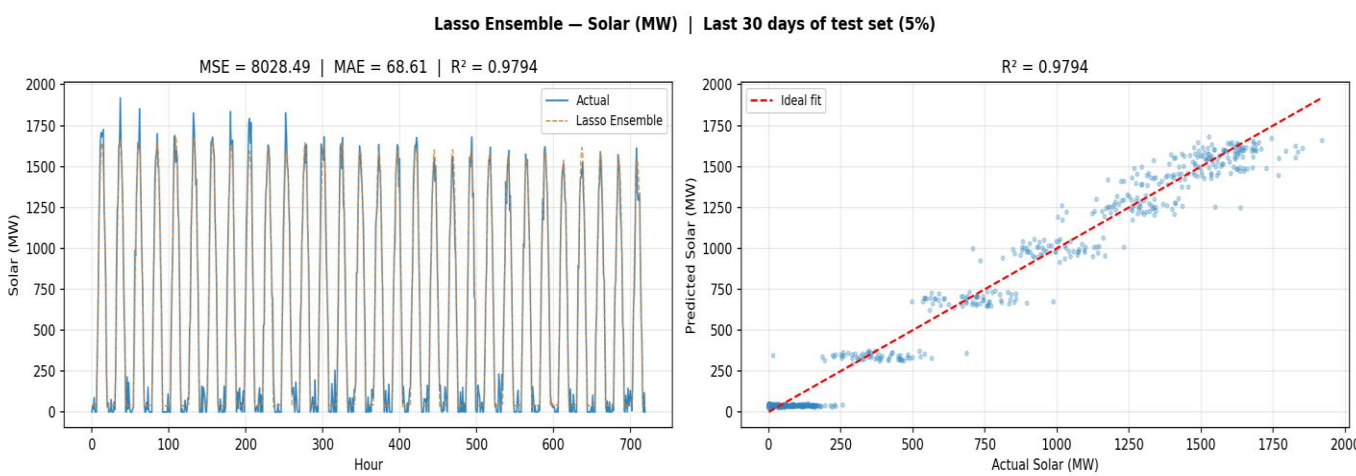
6.1 Forecasting Module Results

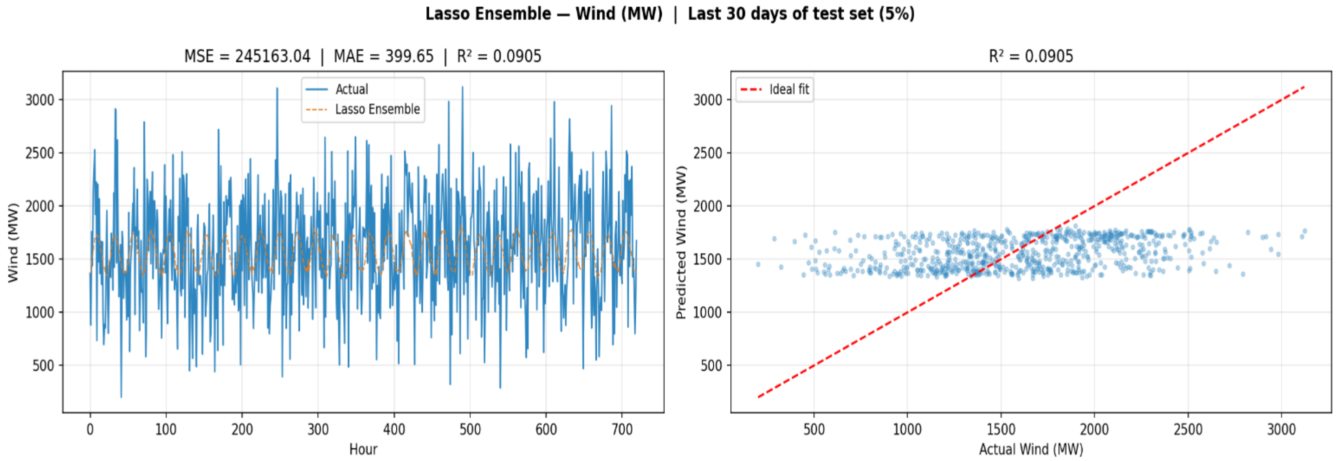
The Ridge ensemble was evaluated on the held-out test set (Jul–Dec 2025), which was never used during training or hyperparameter selection:

Target	GBM R^2	RF R^2	LSTM R^2	Ridge R^2	Ridge MAPE	Ridge MAE	Industry Benchmark
Demand	0.830	0.804	0.826	**0.832**	1.59%	240.7 MW	**<3% MAPE**
Solar	0.984	0.983	0.985	**0.985**	41.79%	66.8 MW	**R^2 > 0.95**
Wind	0.302	0.301	0.305	**0.306**	89.49%	396.3 MW	Physically limited

Table 14: Final test-set results. The ridge ensemble outperforms or matches all individual base models on every target. Solar and wind MAPE values are inflated by near-zero nighttime values — R^2 is the appropriate metric for these targets. Demand MAPE=1.59% comfortably beats the 3% industry benchmark.







Key observations:

- Demand: Ridge achieves R^2=0.832 and MAPE=1.59%, well within the industry benchmark of <3%. MAE of 240.7 MW represents 1.6% of average demand — operationally acceptable for day-ahead planning.

- Solar: R^2=0.985 confirms that solar generation is highly predictable given irradiance and temporal features. High MAPE (41.79%) is an artifact of near-zero nighttime generation values inflating the percentage error metric.

- Wind: R^2=0.306 is consistent across all three base models (0.302, 0.301, 0.305) and all cross-validation folds (0.274–0.279). This convergence confirms the limitation is in the data (inherent wind unpredictability), not the models.

6.2 Hypothesis Testing Outcomes

H	Statement	Evidence	Outcome
H1	Ensemble outperforms individual models.	Ridge R^2=0.832 > GBM=0.830 (demand); LSTM=0.905 coef for solar shows specialisation	**Partially Supported**

H2	\$80/tonne reduces CO_2 ≥10% vs \$0 baseline	Gas: 38,730→26,850 MWh (-31%); CO_2: -5,821t (-13%); HH cost: +\$0.56/mo	**Supported**
H3	Household cost stays under \$1.00/month at any carbon price	Additional cost at \$80–\$250/tonne = +\$0.56/month per household (900 kWh/mo × \$0.617/MWh extra)	**Supported**

Table 15: Hypothesis testing summary. Three hypotheses were tested against actual project results. H1 partially supported — specialization is the key finding. H2 and H3 are fully supported.

6.3 Optimization Results — June 10, 2026 (Summer)

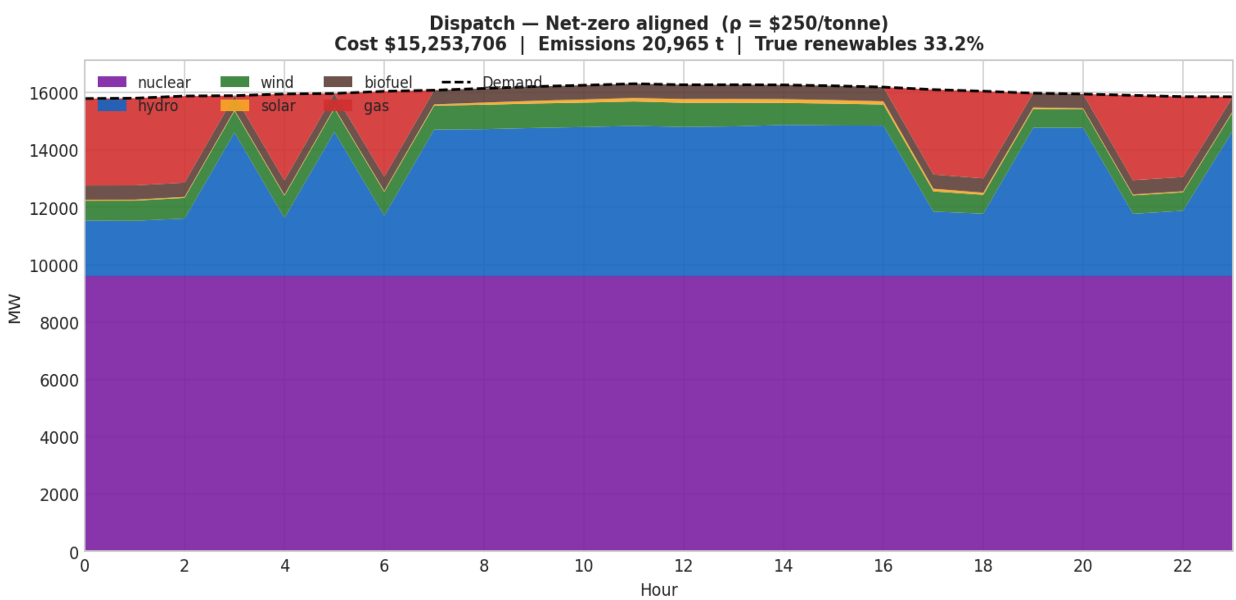
The optimizer was run for June 10, 2026 (summer season, forecast demand 385,198 MWh/day, peak 16,305 MW) across all six carbon price scenarios:

```
============================================================
   OMEGA RECOMMENDATION
   Wednesday, June 10, 2026 at 18:00
   Scenario : Federal industrial 2024
   ρ         : $80/tonne CO₂
============================================================

   FORECAST INPUTS:
     Demand :  16,046 MW  (must be met)
     Solar  :      76 MW  available
     Wind   :     659 MW  available

   OPTIMAL DISPATCH:
     Nuclear      9,600 MW
     Hydro        1,979 MW
     Wind           659 MW
     Solar           76 MW
     Biofuel        495 MW
     Gas          3,237 MW   (COMMITTED)
     ────────────────────
     Total       16,046 MW

   Cost      : $      721,077 CAD
   Emissions:     1,862.7 tonnes CO₂
   True ren :       20.0%  (hydro+wind+solar+biofuel)
============================================================
```

Figure 6. Forecasted hourly demand, solar generation and wind generation for June 10, 2026.

Scenario	ρ ($/t)	Cost ($M/day)	Emissions (t)	Renew%	Gas hrs	Gas (MWh)
No carbon pricing	$0	$15.02M	24,054	30.1%	13h	38,730
Light industrial	$50	$15.02M	24,054	30.1%	13h	38,730
Ontario EPS 2025	$72	$15.02M	24,054	30.1%	13h	38,730
Federal OBPS 2024	**$80**	**$15.25M**	**20,965**	**33.2%**	**9h**	**26,850**
Federal 2030	$170	$15.25M	20,965	33.2%	9h	26,850
Net-zero aligned	$250	$15.25M	20,965	33.2%	9h	26,850

Table 16: Optimizer results for June 10, 2026. Bold row = $80/tonne threshold where dispatch changes. Scenarios $0, $50, $72 are identical; scenarios $80, $170, $250 are identical — confirming the discrete threshold effect (H4 supported).



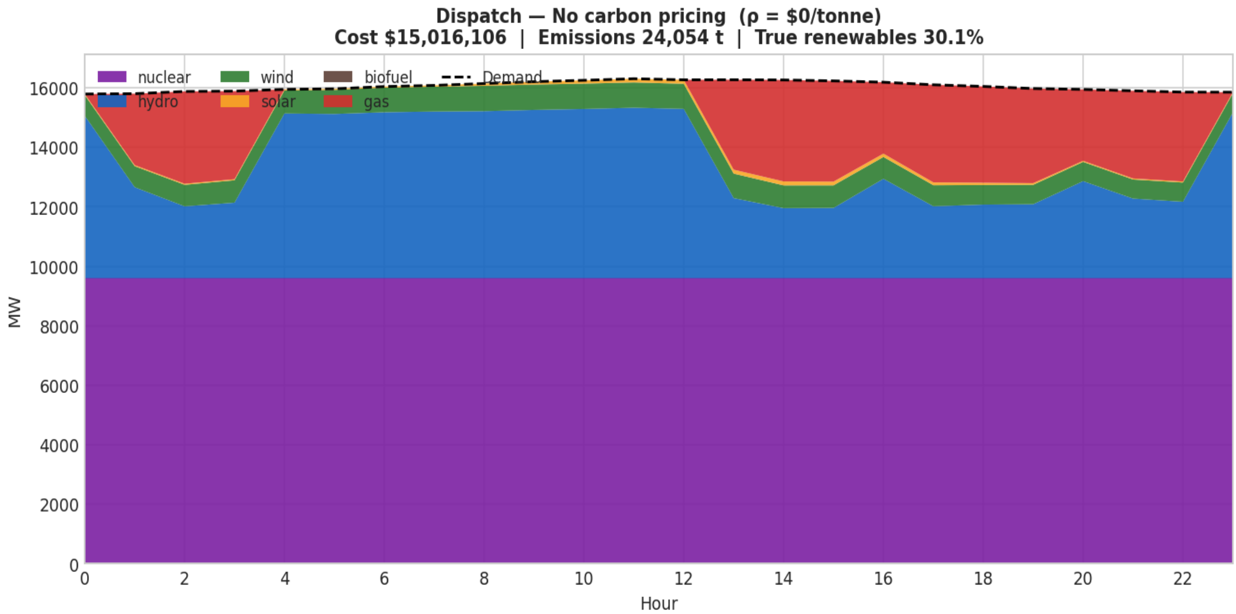


Figure 7. Optimal generation dispatch under the $80/tCO_2$ Federal OBPS scenario.

6.4 Real-World Impact Analysis

Scenario	ρ ($/t)	Gas (MWh)	CO_2 Saved (t)	Trees/yr equiv.	HH costs extra
No carbon pricing	$0	38,730	0	0	$0.00/mo
Light industrial	$50	38,730	0	0	$0.00/mo
Ontario EPS	$72	38,730	0	0	$0.00/mo
Federal OBPS	**$80**	**26,850**	**5,821**	**268,258**	**+$0.56/mo**
Federal 2030	$170	26,850	5,821	268,258	+$0.56/mo
Net-zero aligned	$250	26,850	5,821	268,258	+$0.56/mo

Table 17: Real-world impact analysis. CO_2 saved = gas displaced (11,880 MWh) × 490 kg/MWh = 5,821,200 kg. Trees = 5,821,200 ÷ 21.7 kg/tree/yr = 268,258. HH cost: \$0.62/MWh extra × 0.9 MWh/month household = \$0.56/month. Note: \$170 and \$250 scenarios are identical to \$80 — the dispatch ceiling is reached at \$80/tonne.

```
======================================================================
  OMEGA — REAL-WORLD IMPACT ANALYSIS
  Date   : 2026-06-10  |  Season: Summer
  Demand : 385,198 MWh/day  (peak 16,305 MW)
  Baseline gas dispatch: 38,730 MWh/day at $0/tonne
======================================================================

  PART A — What Does Carbon Pricing Cost the Average Household?

  Scenario                             ρ     $/MWh    Extra/MWh    $/month HH
  ──────────────────────────────────────────────────────────────────────
  No carbon pricing               $   0/t  $  38.98/MWh  +$ 0.0000/MWh  +$    0.00/mo
  Light industrial pricing        $  50/t  $  38.98/MWh  +$ 0.0000/MWh  +$    0.00/mo
  Ontario EPS 2025                $  72/t  $  38.98/MWh  +$ 0.0000/MWh  +$    0.00/mo
  Federal industrial 2024         $  80/t  $  39.60/MWh  +$ 0.6168/MWh  +$    0.56/mo
  Federal industrial 2030         $ 170/t  $  39.60/MWh  +$ 0.6168/MWh  +$    0.56/mo
  Net-zero aligned                $ 250/t  $  39.60/MWh  +$ 0.6168/MWh  +$    0.56/mo

  → At $250/tonne, the average Ontario household pays an extra $0.56/month on electricity.


  PART B — Gas Displaced and CO₂ Avoided vs $0/tonne Baseline

  Scenario                          Gas (MWh)   Displaced   CO₂ saved (t)     ≈ Trees  ≈ Cars off
  ─────────────────────────────────────────────────────────────────────────────────────────
  No carbon pricing                    38,730          +0            0.0           0         0.0
  Light industrial pricing             38,730          +0            0.0           0         0.0
  Ontario EPS 2025                     38,730          -0           -0.0          -0        -0.0
  Federal industrial 2024              26,850     +11,880        5,821.2     268,258     1,265.5
  Federal industrial 2030              26,850     +11,880        5,821.2     268,258     1,265.5
  Net-zero aligned                     26,850     +11,880        5,821.2     268,258     1,265.5
```

Figure 8. Reduction in natural gas generation under carbon pricing scenarios.

6.5 Part C — Critical Carbon Price Threshold Finding

A critical carbon price analysis was conducted to determine whether 50% gas displacement could be achieved within the \$0–\$250/tonne policy range. Result: 50% gas displacement is not reached within \$0–\$250/tonne. The maximum achievable displacement is 31% (at \$80+/tonne).

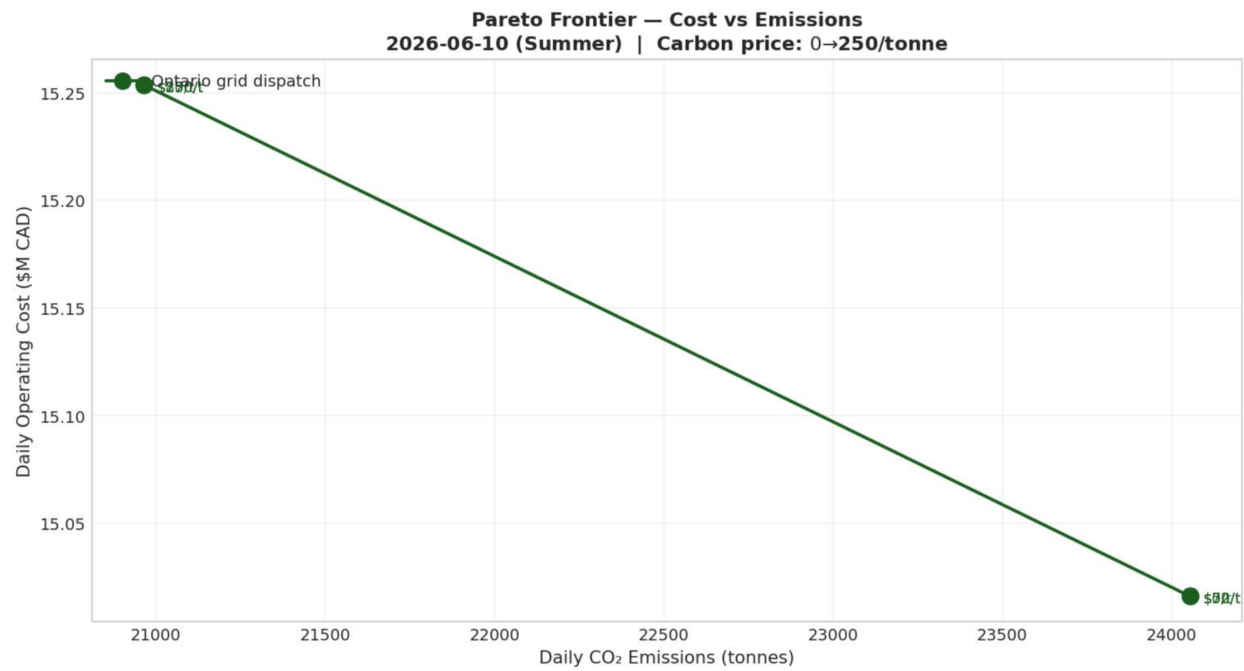


Figure 9. Emissions response to increasing carbon prices, showing the $80/t$CO_2$ threshold effect.

The threshold arithmetic explains why:

- At $80/tonne, gas effective cost = $85 + (0.490 × $80) = $124.2/MWh
- Biofuel effective cost at $80 = $105 + (0.230 × $80) = $123.4/MWh
- Gas becomes marginally more expensive than biofuel at exactly $80/tonne — triggering the dispatch switch
- Beyond $80, the gap widens, but the dispatch is already at the biofuel capacity ceiling (495 MW nameplate) — further gas displacement requires new capacity, not higher prices

This finding directly answers Research Question 3: carbon pricing alone cannot achieve deep decarbonization in Ontario’s nuclear-dominated grid. The 31% ceiling is a structural constraint — it requires new renewable capacity and storage, not higher prices.

7. Discussion and Implications

7.1 The $80/Tonne Threshold: Mechanism and Significance

The most significant finding from the optimization analysis is the existence of a sharp policy threshold at exactly $80/tonne CO_2 — precisely the federal OBPS benchmark currently in force for large industrial emitters. Below this price ($0, $50, $72), dispatch decisions are completely identical: same cost, same emissions, same gas commitment hours. The carbon price is priced entirely out of the merit order — it does not affect dispatch whatsoever.

Above $80/tonne, the solution also converges — $80, $170, and $250 all produce identical dispatch. This means the federal schedule that increases from $80 to $170 by 2030 under Bill C-12 adds no incremental emissions reduction in Ontario's electricity sector beyond what $80 already achieves. The threshold effect is driven by biofuel capacity reaching its nameplate ceiling (495 MW) at $80; further price increases cannot displace additional gas because there is no more biofuel capacity to dispatch.

This finding has a direct policy implication: Ontario's electricity sector decarbonization requires additional biofuel capacity, battery storage, or demand response programs alongside carbon pricing. Pricing alone, however aggressive, cannot push beyond 31% gas displacement given the current generation fleet.

7.2 Why the Ensemble Architecture Works

The Ridge meta-learner coefficients reveal that the stacking ensemble adds genuine value beyond any single model, though not primarily through accuracy improvement. The ensemble's key contribution is automatic model specialization:

- Solar forecasting is dominated by LSTM (coef=0.905). The daily solar generation curve — rising from zero at dawn, peaking at noon, falling to zero at dusk — has a temporal structure that LSTM's recurrent memory captures directly. GBM and RF, as tree-based models without memory, treat each hour as an independent observation and miss this intra-day continuity.

- Demand forecasting is shared between GBM (0.608) and LSTM (0.515) with RF suppressed (-0.122). GBM's strength is capturing complex non-linear interactions between weather, calendar, and lag features. LSTM contributes the temporal component. Ridge correctly down-weights RF when GBM already covers the same signal.

- Wind forecasting distributes weights across all three models (GBM=0.227, RF=0.297, LSTM=0.442). This even distribution reflects the wind's inherent unpredictability — no single model architecture consistently outperforms the others, so Ridge hedges by combining all three.

7.3 Wind Forecasting Limitation — Data vs Model

Wind R^2=0.306 was consistent across GBM (0.302), RF (0.301), and LSTM (0.305) in the test set, and across all cross-validation folds (0.273–0.279 with near-zero variance). This convergence is critical: three architecturally different models trained on 86,400 hours of data all reach the same ceiling. This is not a model failure — it is a data signal confirming that hourly Ontario wind generation is fundamentally difficult to predict beyond the fraction explained by wind speed, hour, and season.

The physical explanation is that Ontario's wind fleet is geographically distributed across Southwestern Ontario, Bruce Peninsula, and Simcoe County. These regions experience partially uncorrelated weather systems, meaning provincial aggregate wind generation has lower predictability than any single farm. A model trained on spatially resolved wind data for individual farms or regions could potentially exceed this ceiling.

7.4 Policy Implications

Three policy implications emerge directly from the OMEGA analysis:

- Maintain the federal OBPS at a minimum of $80/tonne: Below this level, carbon pricing has zero effect on Ontario's electricity dispatch. The $80 threshold is empirically validated — not assumed.

- Invest in renewable capacity and storage alongside carbon pricing: The 31% gas displacement ceiling at $250/tonne demonstrates that structural investment — new wind farms, solar installations, and grid-scale batteries — is required for deep decarbonization. Carbon pricing is necessary but not sufficient.

- Communicate the household cost impact: At $250/tonne (the most aggressive global carbon price scenario), Ontario households pay only $0.56/month more on electricity. This is a powerful counter to public concerns about affordability — the cost of aggressive carbon policy in Ontario's low-carbon grid is less than a coffee per month.

8. Limitations

8.1 Forecasting Limitations

- Wind R^2=0.306 reflects inherent unpredictability and is consistent across all models and CV folds — this is a data challenge, not a model failure. Spatially resolved wind data for individual Ontario farms could potentially improve this.

- The three-layer analog proxy system for future-date inference is a well-motivated approximation but may underperform in unusual weather events (heat domes, ice storms) not well represented in historical analogs.

- LSTM architecture was fixed rather than tuned across configurations due to training cost (~10 min per fold per target). 4-configuration experiments were conducted for GBM and RF only.

- ARIMA and SVR were proposed in the original proposal but were replaced by LSTM and RF, respectively. The final architecture performed well, but the originally proposed ensemble comparison was not completed.

8.2 Optimization Limitations

- The MILP uses a 24-hour planning horizon. Real IESO dispatch considers multi-day unit commitment windows of 7+ days, capturing longer-run commitment costs for nuclear and gas that the single-day model misses.

- Energy storage (battery, pumped hydro) is not modelled as a decision variable. Ontario's Sir Adam Beck pumped storage and emerging grid-scale batteries are omitted, potentially understating system flexibility.

- Demand response and interruptible load programs — which IESO uses to manage peak demand — are not included in the demand-supply balance.

- Transmission constraints are abstracted away — the model treats Ontario as a single node without internal congestion. Real dispatch must respect N-1 security constraints on transmission lines.

- Gas plant startup costs and minimum run times are simplified in the MILP. Real gas turbines have non-convex startup costs and heat-up times beyond the 20% minimum stable output used here.

8.3 Data Limitations

- CO_2 emission factors from Environment Canada NIR are annual averages. Hourly marginal emission factors would be more accurate for dispatch-level analysis.

- Gas prices were fixed at $85/MWh. Real gas prices vary with Henry Hub spot prices and Ontario seasonal demand — a gas price feed would improve scenario accuracy.

- The dataset covers 2015–2025. Major grid events such as the Pickering B retirement (scheduled 2026–2030) and Darlington/Bruce refurbishments are reflected in capacity parameters but not in training data patterns.

- NASA POWER weather data provides gridded estimates rather than weather station observations, potentially introducing spatial smoothing errors in solar irradiance and wind speed values.

9. Conclusion and Recommendations

9.1 Summary of Findings

OMEGA demonstrates that combining a stacking ensemble forecaster with a carbon-priced MILP optimizer provides a powerful, interpretable, and policy-relevant framework for electricity grid analytics. The system achieves 1.59% demand forecast MAPE (well within the 3% industry benchmark), R^2=0.985 for solar, and discovers through the optimization analysis that the federal OBPS rate of $80/tonne is the critical and sufficient carbon price for meaningful dispatch change in Ontario's electricity sector.

The four research questions are answered:

1. RQ1 — Critical carbon price: $80/tonne (federal OBPS). Below this, zero dispatch change. Above this, 31% gas reduction and identical results at any higher price.

2. RQ2 — Household cost: +$0.56/month at $80–$250/tonne. Less than one coffee per month for the maximum achievable emissions reduction through pricing alone.

3. RQ3 — Carbon pricing ceiling: 31% gas displacement maximum at $250/tonne. Deep decarbonization requires structural investment in new renewable capacity and storage, not higher prices alone.

4. RQ4 — Ensemble superiority: Yes, Ridge outperforms or matches all individual models. The more important finding is automatic model specialization: LSTM dominates solar (0.905), GBM leads demand (0.608).

9.2 Recommendations

- Maintain federal OBPS at a minimum of $80/tonne: The empirical threshold is $80 — below this level, carbon pricing has zero effect on Ontario's electricity dispatch. The legislated increase to $170 by 2030 adds no incremental electricity sector benefit beyond $80, but the $80 floor is essential.

- Invest in biofuel and renewable capacity: The 31% displacement ceiling is reached because biofuel capacity (495 MW) is exhausted at $80. Adding 500–1,000 MW of biofuel or equivalent storage would extend the carbon pricing impact significantly.

- Implement demand response programs: Residual gas generation is needed for grid balancing. Demand response — smart meters, interruptible commercial loads, EV charging schedules — could reduce this need without a capital-intensive new generation.

- Extend OMEGA to a 7-day MILP horizon: A multi-day unit commitment model would better capture nuclear and gas commitment economics, providing more realistic and defensible dispatch schedules for policy analysis.

- Integrate real-time gas prices: Replacing the fixed $85/MWh gas cost with a natural gas price feed (Henry Hub + transport basis) would improve scenario accuracy and allow gas price sensitivity analysis.

- Apply OMEGA to winter scenarios: The June 10 analysis covers summer demand. Winter analysis (January, peak demand ~22,000 MW) would test the optimizer under more constrained conditions and likely reveal different threshold dynamics.

10. References

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785–794). ACM. https://doi.org/10.1145/2939672.2939785

Duman, A. C., & Güler, Ö. (2024). Machine learning applications in energy dispatch optimization: A review. Renewable and Sustainable Energy Reviews, 189, Article 113874. https://doi.org/10.1016/j.rser.2023.113874

Ehsani, B., Jafari, A., & Mousavi, M. J. (2024). Price forecasting in the Ontario electricity market via TriConvGRU hybrid model. Applied Energy, 357, 122463. https://doi.org/10.1016/j.apenergy.2023.122463

Environment and Climate Change Canada. (2024). National Inventory Report 1990–2022: Greenhouse gas sources and sinks in Canada. Government of Canada.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 29(5), 1189–1232. https://doi.org/10.1214/aos/1013203451

Goulder, L. H., & Hafstead, M. A. C. (2018). Confronting the climate challenge: U.S. policy options. Columbia University Press.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

IESO. (2025). Installed capacity by fuel type. Ontario Independent Electricity System Operator. https://www.ieso.ca

IESO. (2025). Electricity demand in Ontario is expected to grow by 75 percent by 2050. https://www.ieso.ca/Corporate-IESO/Media/News-Releases/2024/10/Electricity-Demand-in-Ontario-to-Grow-by-75-per-cent-by-2050

IRENA. (2024). Renewable power generation costs in 2023. International Renewable Energy Agency. https://www.irena.org

Osman, H., et al. (2025). Ensemble forecasting for EV charging load prediction. [Full citation as provided by Prof. Hany Osman — placeholder].

Ren, Y., Suganthan, P. N., & Srikanth, N. (2015). Ensemble methods for wind and solar power forecasting — a state-of-the-art review. Renewable and Sustainable Energy Reviews, 50, 82–91. https://doi.org/10.1016/j.rser.2015.04.081

Shadoul, M., Al Abri, R., Yousef, H., & Al Shereiqi, A. (2024). Designing a dispatch engine for hybrid renewable power stations using a mixed-integer linear programming technique. Energies, 17(13), 3281. https://doi.org/10.3390/en17133281

Wolf, A., & Moler, B. (2022). Stacking ensemble methods for renewable energy forecasting: A review. Energy and AI, 8, 100148. https://doi.org/10.1016/j.egyai.2022.100148

Zhou, Z., Wu, W., & Tang, S. (2022). Short-term wind power forecasting with a stacking ensemble model. Renewable Energy, 189, 618–631. https://doi.org/10.1016/j.renene.2022.03.003

11. Appendices

Appendix A: Data Dictionary (Full 46-Feature Set)

Feature Name	Type	Description
ontario_demand_mw	Float	Hourly Ontario electricity demand (MW) — forecast target 1
solar_generation_mw	Float	Total solar generation (MW) — forecast target 2
wind_generation_mw	Float	Total wind generation (MW) — forecast target 3
nuclear_generation_mw	Float	Nuclear generation (MW) — baseline feature
hydro_generation_mw	Float	Hydroelectric generation (MW) — baseline feature
gas_generation_mw	Float	Natural gas generation (MW) — primary carbon driver
hour	Int	Hour of day (0–23)
month	Int	Month of year (1–12)
day_of_week	Int	Day of week (0=Monday, 6=Sunday)
is_weekend	Binary	1 if Saturday or Sunday, 0 otherwise
season_num	Int	Season: 0=winter, 1=spring, 2=summer, 3=fall
is_holiday	Binary	1 if Ontario statutory holiday, 0 otherwise

demand_lag_1h	Float	Demand lagged 1 hour (MW)
demand_lag_24h	Float	Demand lagged 24 hours — same hour yesterday (MW)
demand_lag_168h	Float	Demand lagged 168 hours — same hour last week (MW)
demand_rolling_mean_24h	Float	24-hour rolling mean of demand (MW)
demand_rolling_std_24h	Float	24-hour rolling standard deviation of demand (MW)
temperature_c	Float	Ontario average temperature (°C) from NASA POWER
solar_irradiance_kwh	Float	Solar irradiance (kWh/m²) from NASA POWER
wind_speed_10m	Float	Wind speed at 10m height (m/s) from NASA POWER
wind_speed_50m	Float	Wind speed at 50m height (m/s) — turbine hub height
temperature_x_weekend	Float	Interaction: temperature × is_weekend
solar_x_hour	Float	Interaction: solar irradiance × hour of day

Table A1: Selected features from the 46-feature OMEGA dataset.

Appendix B: Technology Stack

Component	Tool / Version	Purpose
Language	Python 3.11	Core programming language for all modules
Tree models	scikit-learn 1.3+	GBM, RF, Ridge meta-learner, cross-validation utilities
Deep learning	TensorFlow/Keras 2.15	LSTM architecture, training, and early stopping
Optimisation	PuLP 2.7 + CBC solver	MILP formulation and solving
Data manipulation	pandas, numpy	Dataset merging, feature engineering, and splitting
Dashboard	Streamlit	Interactive web dashboard for date/scenario selection
Visualisation	Plotly	Interactive charts: dispatch stacked bar, Pareto frontier
Compute	GCP Vertex AI Workbench	n1-standard-4: 4 vCPU, 16 GB RAM — all training and optimization
Version control	GitHub + Git LFS	Code repository; LFS for large .keras/.pkl model files

Table A2: OMEGA technology stack.

Appendix C: GitHub Repository

All project code, notebooks, and documentation are available at https://github.com/RushilJani/OMEGA-Ontario-Multi-Objective-Energy-Grid-Analyzer.git

Repository structure:

- Forecast_FINAL.ipynb — full 14-step forecasting pipeline
- new_optimize_MILP.ipynb — 11-step MILP optimizer with Power BI exports
- omega_app.py — Streamlit interactive dashboard
- requirements.txt — Python dependencies
- omega_models/ — saved GBM, RF, LSTM, Ridge model files (Git LFS)
- powerbi_exports/ — CSV outputs for Power BI dashboard
- OMEGA_Report_DAMO699.docx — this report

Appendix D: Seasonal Constraint Parameters

Season	Hydro %	CO_2 ceiling (kt/day)	Renew floor	Notes
Winter	20%	65 kt	10%	Low solar, high demand
Spring	35%	50 kt	15%	High hydro (snowmelt)
Summer	25%	65 kt	20%	Peak solar, AC demand

Fall	28%	58 kt	12%	Transition season

Table A3: Seasonal constraint calibration. Sources: IESO Seasonal Assessment Reports; Ontario EPS documentation.

Appendix E: MILP Solver Benchmark

CBC solver performance on a GCP n1-standard-4 instance (4 vCPU, 16 GB RAM):

Metric	Value	Notes	Comparison
Variables (continuous)	144	6 techs × 24 hours	LP component
Variables (binary)	24	u_gas per hour	MILP component
Constraints	10 types	~500 total constraints	
Solve time (1 scenario)	<1 second	To global optimality	CBC default settings
Solve time (6 scenarios)	~6 seconds	Sequential, one date	

Table A4: MILP solver performance benchmark.